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Editors-in-Chief
Smart Agricultural Technology

Dear Editors,

Please consider the manuscript, "From Candidate Scores to Plant Recovery in Finite Crop–Weed Review Queues," for publication in *Smart Agricultural Technology*.

The study addresses a practical question for agricultural vision: how many distinct plants remain available when candidate formation, role qualification, ranking, and a finite review capacity are evaluated as one chain? We answer this question on the official WeedsGalore spatial split and in a separately locked CWFID transfer test.

The framework separates spatial proposal recall, role-qualified recall, and deterministic one-to-one plant recovery. A paired 471-candidate comparison isolates proposal commissioning from additional review burden. Image-cluster bootstrap intervals quantify the resulting gain, and standard candidate AUC and average precision are retained as diagnostics rather than end-to-end endpoints.

The model comparison uses matched target supervision and the same fixed $K = 20$ contract. It includes a ResNet-18 U-Net, DeepLabV3+, Mask R-CNN, frozen DINOv2 rankers, and semantic probability or entropy ordering. Mask R-CNN provides the strongest internal one-to-one recovery, while a simple validation-selected semantic score remains competitive. Two independent operators also reviewed a frozen 518-region queue and achieved 94.2% exact agreement on weed reviewability ($\kappa = 0.893$).

The external CWFID protocol was frozen before image or annotation access and opened once. All five WeedsGalore-selected checkpoints generated zero CWFID candidates at their fixed operating points. This negative transfer result is central to the paper because it separates internal optimisation stability from acquisition transfer and identifies component calibration as an explicit deployment boundary.

The accompanying package includes editable figures, source-data tables, exact matching outputs, frozen configurations, model and data hashes, failed-run records, and clean-room replay commands. The manuscript is original, is not under consideration elsewhere, and has been approved by all authors.

Sincerely,

Zhuo Chen
on behalf of all authors